

Ali Rahimi-Kalahroudi

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EDUCATION

Université de Montréal

M.Sc. in Computer Science
Montréal, Canada

Sharif University of Technology

B.Sc. in Computer Science
Tehran, Iran

SKILLS

Programming

Skilled in • Python • Scala
Familiar with • C++ • Java

ML Tools

Skilled in • PyTorch • NumPy
Familiar with • Tensorflow • JAX

Technologies

- **Data-related:** – Apache Spark
– PySpark – Apache Hadoop
- **Misc:** – Docker – Singularity

HONORS AND AWARDS

- **Bourse D'excellence DIRO**
Université de Montréal
Montréal, Canada | 2022
- **M.Sc. Tuition Exemption Scholarships (Bourse C)**
Université de Montréal
Montréal, Canada | 2021
- **International Undergraduate Excellence Award**
Chair for CAMPAR of TUM
Munich, Germany | 2019
- **Silver Medal**
24th Iranian National Olympiad in Informatics(INOI)
Tehran, Iran | 2014

PUBLICATIONS

- *Replay Buffer with Local Forgetting for Adaptive Deep Model-Based Reinforcement Learning*
Rahimi-Kalahroudi, A., Rajendran, J., Momennejad, I., Van Seijen, H., and Chandar, S.
Deep Reinforcement Learning Workshop, **NeurIPS**, 2022 (paper)
 - *Towards Evaluating Adaptivity of Model-Based Reinforcement Learning Methods*
Wan, Y.*, **Rahimi-Kalahroudi, A.***, Rajendran, J., Momennejad, I., Chandar, S., and Van Seijen, H.
International Conference on Machine Learning (**ICML**), 2022 (paper, code)
- * denotes equal contribution.

EXPERIENCE

Mila - Quebec AI Institute | Research M.Sc. Student

Jan 2021 – Present | Montréal, Canada

- Supervisor: Prof. Sarath Chandar
- Focusing on fundamental questions at the intersection of model-based reinforcement learning (MBRL) and continual learning, specifically, highlighting the importance of local adaptivity in MBRL, identifying design choices in popular methods that hinder it, and proposing locally adaptive MBRL methods.

iCAD Dental | ML Research Developer (part-time)

Jan. 2022 – May. 2022 | Ottawa, Canada

- Designed a pipeline for the classification and clustering of dental objects, specifically crowns, utilizing a database of SSM files as the primary source of data.

CafeBazaar Inc. | Data Scientist

Sep. 2017 – Jan. 2021 | Tehran, Iran

- CafeBazaar is Iran's third-party android marketplace with more than 40 million users.
- Notable contributions:
 - As a member of the Personalization team, developed and implemented the app recommender system model trainer using the implicit-als algorithm, conducted A/B testing to validate improvements, and devised a solution to diversify recommended app categories.
 - As a member of the Search-ads team, led the data science, mechanism design, and optimization efforts of a team of two data scientists and one intern. Responsibilities included designing the auction mechanism and systems for setting the reserved price, profiling users, and suggesting relevant query keywords to developers. Through these efforts, the Search-ads team became one of the top three daily revenue generators for the company. (Towards Data Science post)

Chair for CAMPAR | Undergraduate Research Assistant

Jul. 2019 – Oct. 2019 | Munich, Germany

- Contributed to a project focused on 3D camera re-localization in dynamic indoor environments. Coded a valid pose generator, utilizing rendered scenes, for the 3RScan data.

PROJECTS

RLHive

A comprehensive research framework for reinforcement learning, featuring a collection of popular single-agent and multi-agent algorithms implemented in PyTorch. (code, docs)

Image Style Transfer

PyTorch implementation of image style transfer using CNNs. (code)